

Question:

To design and implement a bank account management system in java using classes and objects,demonstrating object-oriented programming concepts such as classes,objects,constructors ,inheritance,method overriding ,deposits,withdrawals,interest calculation and fund transfers.

System Requirements:

Hardware Requirements:

1. Standard PC/Laptop
2. Intel Core i3/i5 processor or equivalent
3. Minimum 4 GB RAM
4. Minimum 500 MB free storage

Software Requirements:

1. Operating System: Windows / Linux / macOS
2. Java Development Kit (JDK 17 or JDK 21)
3. IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
4. Command Prompt / Terminal

Execution Steps:

Step 1: Start the program.

Step 2: Create a SavingsAccount object.

Account Number = 101

Holder Name = Ravi

Balance = 10000

Interest Rate = 5%

Step 3: Create a CurrentAccount object.

Account Number = 102

Holder Name = Rani

Balance = 5000

Overdraft Limit = 2000

Step 4: Display both account details.

Step 5: Deposit 2000 into the savings account.

Balance = 12000

Step 6: Withdraw 1000 from the savings account.

Balance = 11000

Step 7: Calculate 5% interest.

Interest = 550

Balance = 11550

Step 8: Deposit 3000 into the current account.

Balance = 8000

Step 9: Withdraw 7000 from the current account.

Balance = 1000

Step 10: Transfer 2000 from savings to current.

Savings Balance = 9550

Current Balance = 3000

Step 11: Display final account details.

Step 12: Stop the program.

Algorithm:

START

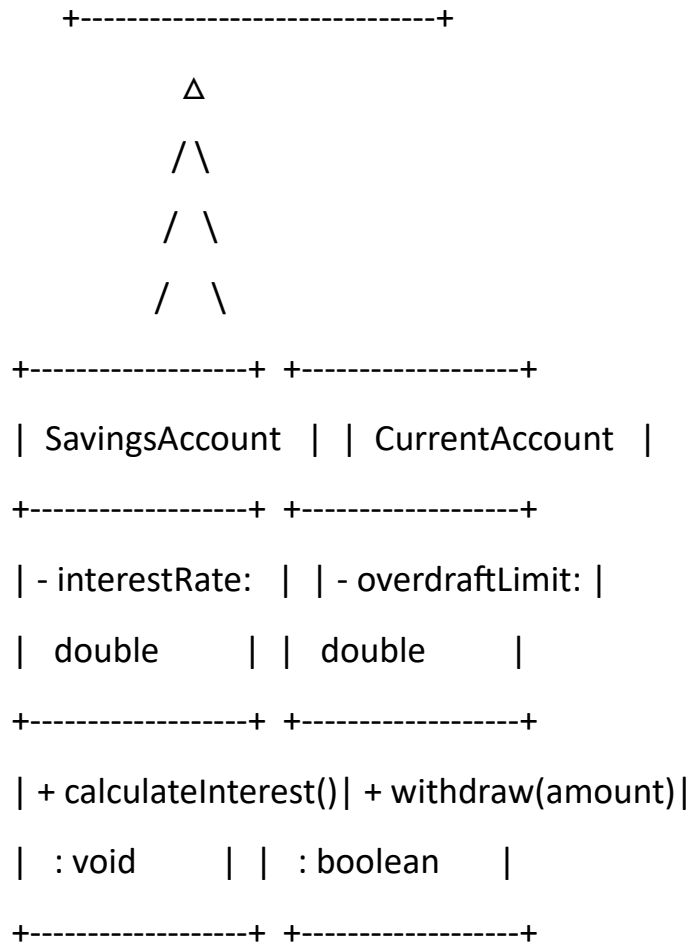
1. Create the Account base class with account details.
2. Define deposit(), withdraw(), transfer(), and displayAccountDetails() methods.
3. Create SavingsAccount class that inherits Account.
4. Define calculateInterest() in SavingsAccount.
5. Create CurrentAccount class that inherits Account.
6. Override withdraw() to allow overdraft.
7. Create SavingsAccount and CurrentAccount objects.
8. Display account details before transactions.

9. Perform deposit, withdrawal, and interest calculation for the savings account.
10. Perform deposit and withdrawal for the current account.
11. Transfer money from savings to current account.
12. Display account details and final balances.

STOP

UML Class Diagram:

```
+-----+
|      Account      |
+-----+
| - accountNumber: int    |
| - holderName: String    |
| - balance: double       |
| - accountType: String   |
+-----+
| + deposit(amount): void    |
| + withdraw(amount): boolean |
| + transfer(receiver, amount) |
| + displayAccountDetails():void |
```



Code:

```
package javacore;
```

```
class Account {
```

```
    int accountNumber;
```

```
    String holderName;
```

```
    double balance;
```

```
    String accountType;
```

```
    Account(int number, String name, double balance, String type) {
```

```
accountNumber = number;  
holderName = name;  
this.balance = balance;  
accountType = type;  
}
```

```
void deposit(double amount) {  
    if (amount > 0) {  
        balance += amount;  
        System.out.println("Deposit successful");  
    } else {  
        System.out.println("Invalid deposit amount");  
    }  
}
```

```
boolean withdraw(double amount) {  
    if (amount > 0 && balance >= amount) {  
        balance -= amount;  
        System.out.println("Withdrawal successful");  
        return true;  
    }  
}
```

```
System.out.println("Insufficient balance");  
return false;  
}
```

```
void transfer(Account receiver, double amount) {  
    if (receiver == null || receiver == this) {  
        System.out.println("Invalid receiver");  
        return;  
    }  

```

```
    if (amount <= 0) {  
        System.out.println("Invalid transfer amount");  
        return;  
    }  

```

```
    if (withdraw(amount)) {  
        receiver.deposit(amount);  
        System.out.println("Transfer successful");  
    }  
}
```

```
void displayAccountDetails() {  
    System.out.println("Account Number: " + accountNumber);  
    System.out.println("Holder Name: " + holderName);  
    System.out.println("Account Type: " + accountType);  
    System.out.println("Balance: " + balance);  
}  
}
```

```
class SavingsAccount extends Account {
```

```
double interestRate;
```

```
SavingsAccount(int number, String name, double balance,  
               double rate) {  
    super(number, name, balance, "Savings");  
    interestRate = rate;  
}
```

```
void calculateInterest() {  
    double interest = balance * interestRate / 100;  
    balance += interest;  
    System.out.println("Interest: " + interest);  
}  
}
```

```
class CurrentAccount extends Account {  
    double overdraftLimit;
```

```
    CurrentAccount(int number, String name, double balance,  
                   double limit) {  
        super(number, name, balance, "Current");  
        overdraftLimit = limit;  
    }
```

```
@Override
```

```
boolean withdraw(double amount) {
```

```
        if (amount > 0 && balance + overdraftLimit >= amount) {  
            balance -= amount;  
            System.out.println("Withdrawal successful");  
            return true;  
        }  
  
        System.out.println("Overdraft limit exceeded");  
        return false;  
    }  
}
```

```
public class BankAccountDemo {  
    public static void main(String[] args) {  
  
        SavingsAccount s = new SavingsAccount(  
            101, "Ravi", 10000, 5);  
  
        CurrentAccount c = new CurrentAccount(  
            102, "Rani", 5000, 2000);  
  
        System.out.println("Before Transactions");  
        s.displayAccountDetails();  
        c.displayAccountDetails();  
  
        System.out.println("\nTransactions");  
    }  
}
```

```
s.deposit(2000);  
s.withdraw(1000);  
s.calculateInterest();  
  
c.deposit(3000);  
c.withdraw(7000);  
  
s.transfer(c, 2000);  
  
System.out.println("\nAfter Transactions");  
s.displayAccountDetails();  
c.displayAccountDetails();  
}  
}
```

Output:

Before Transactions

Account Number: 101

Holder Name: Suresh

Account Type: Savings

Balance: 10000.0

Account Number: 102

Holder Name: Ramesh

Account Type: Current

Balance: 5000.0

Transactions

Deposit successful

Withdrawal successful

Interest: 550.0

Deposit successful

Withdrawal successful

Withdrawal successful

Deposit successful

Transfer successful

After Transactions

Account Number: 101

Holder Name: Suresh

Account Type: Savings

Balance: 9550.0

Account Number: 102

Holder Name: Ramesh

Account Type: Current

Balance: 8000.0